Sangjun (Samuel) Ko

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Work Experience

Kimberly-Clark — North America Family Care R&D Neenah, Wisconsin
Statistician (contract assignment through Staffmark Group) June 2025 – Present

- Serve as statistical lead for Test Method Validation (TMV), Measurement System Analysis (MSA), and experimental design across North America Family Care R&D, partnering with quality, product engineering, and laboratory teams to qualify critical-to-quality (CTQ) test methods.
- Rebuilt Kimberly-Clark's core diaper-leakage modeling engine from legacy SAS into a containerized Python/Flask web app; resolved whole-plot vs. sub-plot error structure partitioning in split-plot repeated-measures trials, achieving benchmark parity across 9 historical studies and reducing report generation from days to minutes for 50+ researchers.
- Designed balanced factorial equivalence studies ($N = 96$) to qualify replacement abrasive mandrels across two instrument lines; detected a persistent 0.7–1.3 mg negative measurement bias that disproved vendor equivalence, halting premature hardware adoption and driving supplier re-machining.
- Engineered an automated time-series data reduction pipeline in Python for transient sensor curves; modeled moisture dissipation dynamics to identify a 15–17 minute plateau, establishing an empirical 14–15 minute reading rule that eliminated subjective technician timing cutoffs.
- Designed multi-factor screening DOEs (including 96-run cart comparison and 48-run fluid insult rate studies) to correlate accelerated benchtop mannequin testing with 3-month consumer home-use test (HUT) data, isolating key design drivers and screening prototypes in weeks rather than months.
- Architected a 13-claim attribute Test Method Validation protocol across two feminine care product lines; designed a multi-round diagnostic framework with blinded sample routing and Cohen's/Fleiss' Kappa agreement analysis to isolate technician misclassifications and lighting variability prior to formal qualification.
- Designed Test Method Characterization (TMC) and formal Gage R&R protocols for destructive fastening tab-tear strength; decomposed variance components under single-pull constraints and established statistical guardbands against consumer-perceived detachment thresholds.
- Formulated technician-blocked Gage R&R designs for tensile strength and speed-dried basis weight under severe laboratory throughput constraints, reducing testing labor by approximately 3× compared to full interleaving while maintaining statistical power to detect measurement noise.
- Conducted retrospective validation on multi-variant adhesive weight data; separated auditor measurement variability from true process variation using statistical tolerance intervals to establish variant-specific pass/fail release specifications.
- Deployed six production web modules on Kimberly-Clark's internal cloud computing hub, including an interactive predictive-model explorer (19 laboratory predictors → 8 consumer attributes), an automated Gage R&R sample-size calculator, and a centralized laboratory test-run database.
- Operationalized multivariate product-performance models into self-service web applications, enabling non-statistical researchers to simulate design trade-offs, explore parameter spaces, and evaluate response surfaces interactively.
- Formulated study designs for multi-site interlaboratory method transfers spanning U.S. and international testing facilities, evaluating repeatability, reproducibility, lab-to-lab relative bias, and specimen variability under practical execution constraints.
- Translate complex statistical findings into actionable engineering recommendations for test-method design, equipment qualification, and root-cause investigations, delivering decision-ready packages to quality and R&D leadership.

STAT 107: Data Science Discovery, University of Illinois Urbana-Champaign Champaign, Illinois
Undergraduate Course Assistant August 2023 – May 2025

- Mentored 300+ students in foundational data-science concepts through interactive laboratory sessions and individualized technical support.
- Managed Git-based updates and created, edited, and proofread instructional content for a learning platform serving 1,000+ students.
- Collaborated with a staff of 100+ course assistants and two professors through weekly planning and coordination meetings.

STAT 443: Professional Statistics, University of Illinois Urbana-Champaign Champaign, Illinois
Undergraduate Course Assistant January 2025 – May 2025

- Graded assignments and provided feedback to 20+ students on statistical analysis, consulting, and communication.
- Evaluated 5+ student presentations for analytical quality and alignment with project expectations.
- Acted as a client in mock consulting projects, posing realistic technical questions and business constraints.

Applied Technologies for Learning in Arts & Sciences Champaign, Illinois
AI/ML Intern Team Leader January 2024 – December 2024

- Led a team of five interns and coordinated priorities, deliverables, and progress with three managers through weekly meetings.
- Developed and evaluated Python machine-learning models using datasets containing 100,000+ records.
- Applied regression, random forests, XGBoost, neural networks, BERT, and transformer-based methods to predictive and natural-language-processing projects.

HappySneeze Remote
Market Research Intern August 2024 – September 2024

- Created an interview guide and conducted 10+ consumer interviews concerning menopause and pelvic-floor health needs.
- Conducted market and consumer-trend research and synthesized findings into product and market recommendations for stakeholders.

KM Internacional S.A. Managua, Nicaragua
Intern May 2023 – August 2023

- Oversaw inventory management and streamlined administrative processes for a small textile company.
- Led three employees in improving record-keeping accuracy to 98% and reducing operational costs by 15%.

Korean Augmentation to the United States Army (KATUSA) Gyeonggi-do, South Korea
Soldier — Infantry and S-4 Logistics June 2021 – December 2022

- Completed an 18-month assignment in bilingual and cross-cultural U.S.–Korean military environments.
- Performed infantry responsibilities involving maintenance and operation of an M1 Abrams tank with the 2nd Squadron, 13th Cavalry Regiment.
- Supported S-4 logistics and inventory operations for the 176th Financial Management Support Unit, maintaining accurate and efficient accountability processes.
- Served as a Korean–English liaison and translator for 2,000+ U.S. and Korean soldiers during joint training exercises and assisted newly stationed personnel with local orientation.

Istmo Textiles S.A. Masaya, Nicaragua
Intern February 2021 – March 2021

- Studied corporate social responsibility, human rights, labor and environmental requirements, industrial safety, and textile import/export operations between Central and North America.
- Participated in community-support activities providing school supplies and food to schools and churches in Masaya.

Projects and Awards

Graduation Prediction Model September 2024 – December 2024

- Increased model AUC from 0.65 to 0.82 through feature engineering and outlier review across 50,000+ student records.
- Built a Streamlit interface using 12 key predictors to provide real-time graduation predictions for university administrators.

3rd Place, 2024 COVID Information Commons Student Paper Challenge September 2024

- Analyzed socioeconomic and mental-health effects of COVID-19 in Mexico using statistical and geospatial methods and proposed interventions for vulnerable groups.

Paraphrase Detection July 2024

- Achieved 90%+ accuracy on the MRPC dataset using BERT, RoBERTa, and SBERT and developed a Gradio interface for real-time evaluation.

International Student Forecasting April 2024

- Processed 100,000+ enrollment records, identified trends, and forecast international enrollment to support resource-allocation recommendations.

Titanic Machine-Learning Model February 2024

- Developed an 80%-accuracy Python classification model and used feature importance to identify key survival factors.

Education

University of Illinois Urbana-Champaign Champaign, Illinois
Bachelor of Science in Statistics, Mathematics Minor; GPA: 3.93/4.00 January 2023 – May 2025

- Relevant coursework: Applied Bayesian Analysis, Advanced Data Analysis, Statistical Modeling, Statistical Data Management, Statistical Consulting, Probability, Real Analysis, Linear Algebra, and Data Science Programming.

University of Michigan–Shanghai Jiao Tong University Joint Institute Shanghai, China
Coursework toward a B.E. in Mechanical Engineering; transferred to UIUC September 2019 – December 2020

- First Class Scholarship.

Skills

Statistical methods:	DOE, TMV, Gauge R&R/MSA, mixed models, ANOVA, multiple comparisons, regression, attribute agreement, bias and precision analysis, equipment equivalency, method transfer
Programming:	Python, R, SQL, JavaScript, HTML/CSS, Excel/VBA
Libraries/frameworks:	Pandas, NumPy, SciPy, statsmodels, scikit-learn, TensorFlow, Flask, Streamlit, Gradio, Matplotlib, Seaborn, Tidyverse, ggplot2
Tools:	JMP, Git, Docker, SAS, Tableau, Microsoft Office
Languages:	Korean and Spanish (native); English (fluent); French and Chinese (basic)
Certificates:	Data Science DISCOVERY; Data Science Certificate; Coursera Business Analytics Certificate

Leadership, Service, and Activities

Volunteer, Habitat for Humanity — Tornado Recovery Menasha, Wisconsin
August 2026

- Cleared storm debris and checked in with households in the Menasha area that requested tornado-recovery assistance through 211 or Habitat for Humanity.

Volunteer, Appleton Public Library Appleton, Wisconsin
August 2025 – Present

- Shelve and organize materials and support "Game On" programs through setup, participant guidance, and activity assistance.

Consultant, Illinois Business Consulting Champaign, Illinois
September 2024 – December 2024

- Collaborated with an interdisciplinary team on market analysis, operational optimization, and growth-strategy recommendations for a client engagement.

Treasurer, Performance UIUC Touch Champaign, Illinois
February 2023 – May 2024

- Planned and executed 10 fundraising initiatives, increasing available funds by 30% to support activities, costumes, and travel.
- Participated in weekly practices and public performances and helped expand the team's social-media audience by 100+ followers.

President, Asian Culture Club November 2018 – March 2019

- Led five students in promoting multicultural engagement at an international high school and organized weekly meetings and monthly activities.

Volunteer, Habitat for Humanity Nicaragua
September 2016 – June 2017

- Supported home and sanitation-system construction and maintenance projects serving low-income communities, contributing to 15 homes.

Teacher, Nanum School Masaya, Nicaragua
August 2016 – December 2016

- Taught English and mathematics to children ages 5–12 through a local community school organized by a Korean Presbyterian church.

Additional Honors

- Army Achievement Medal, United States Army, December 2022.
- First Class Scholarship, University of Michigan–Shanghai Jiao Tong University Joint Institute, 2019.
- Salutatorian and Mathematics Award, Notre Dame International School, 2019.